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FAIR Peer Review and Publication

Fostering a Transparent Publication Ecosystem

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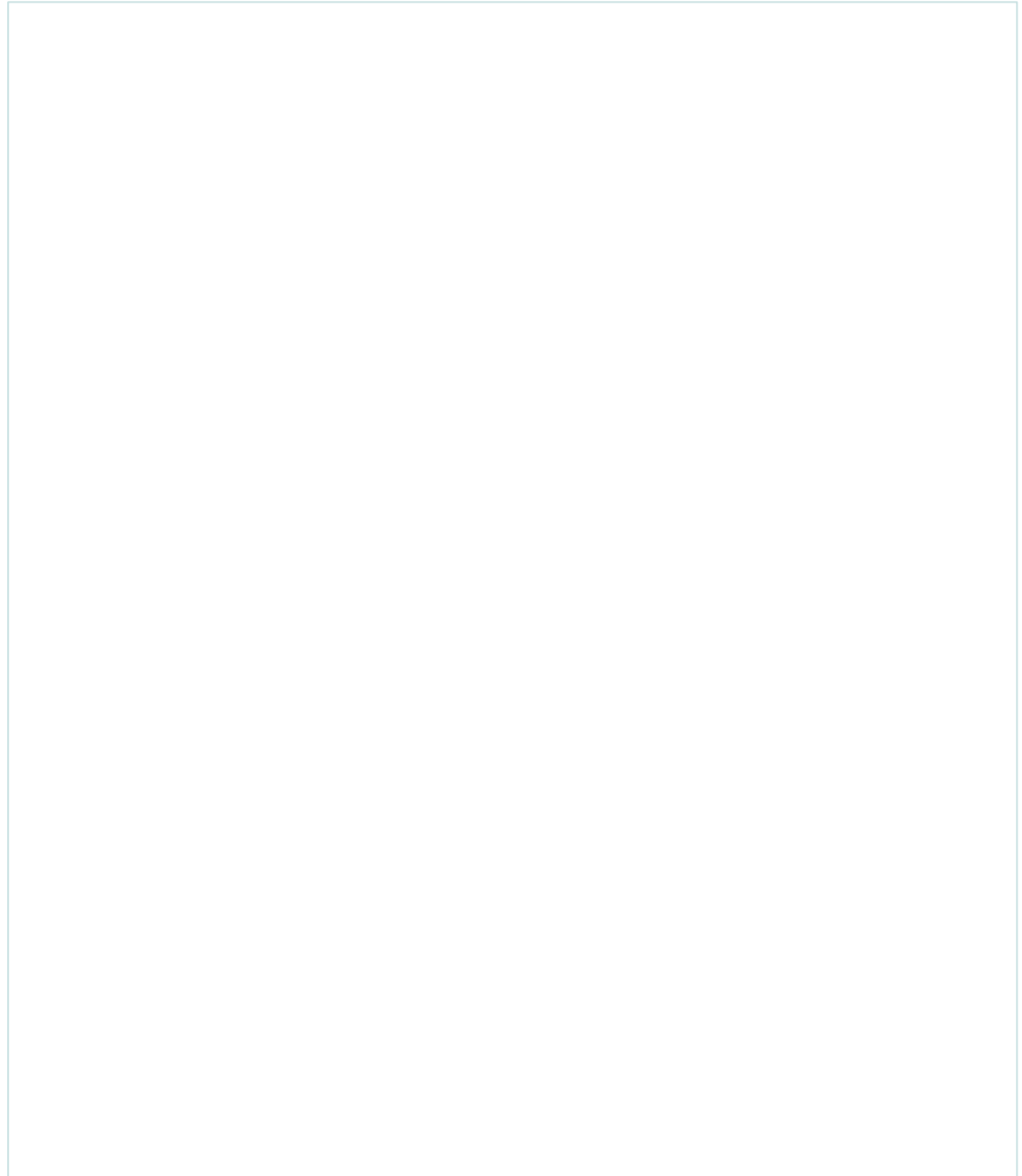
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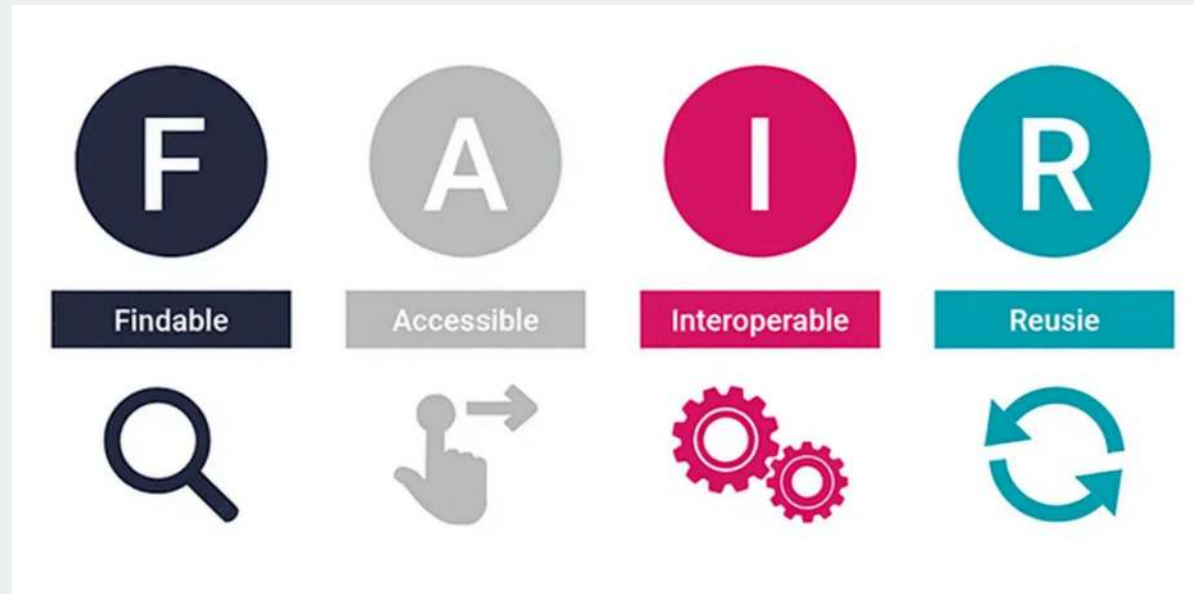
Senior editor: J Biogeography | Ecology and Evolution

The Challenge in Ecology

The complexity of ecological data is immense. Our current publication system can be slow, opaque, and siloed. How can we improve the review and dissemination of vital ecological research?



What are the FAIR Principles?



Findable

Data and metadata easily findable by humans and machine

Accessible

Recoverable by standardized and open protocols

Interoperable

Integration with other data, applications or workflows

Reusable

Well described for replication and/or combination with different contexts

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F - Findable

For data, publications, and reviews to be easily discovered by the Community:

- Globally unique and persistent identifiers (e.g., DOIs for papers, data, and reviews)
- Rich metadata describing the research context



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A - Accessible

Ensuring that research outputs are retrievable:

- Open access publications
- Data and code available in public repositories
- Clear and transparent access conditions



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I - Interoperable

The ability to integrate and work with other data and systems:

- Use of standardized vocabularies and formats for data
- Machine-readable data and metadata
- Fosters large-scale ecological synthesis

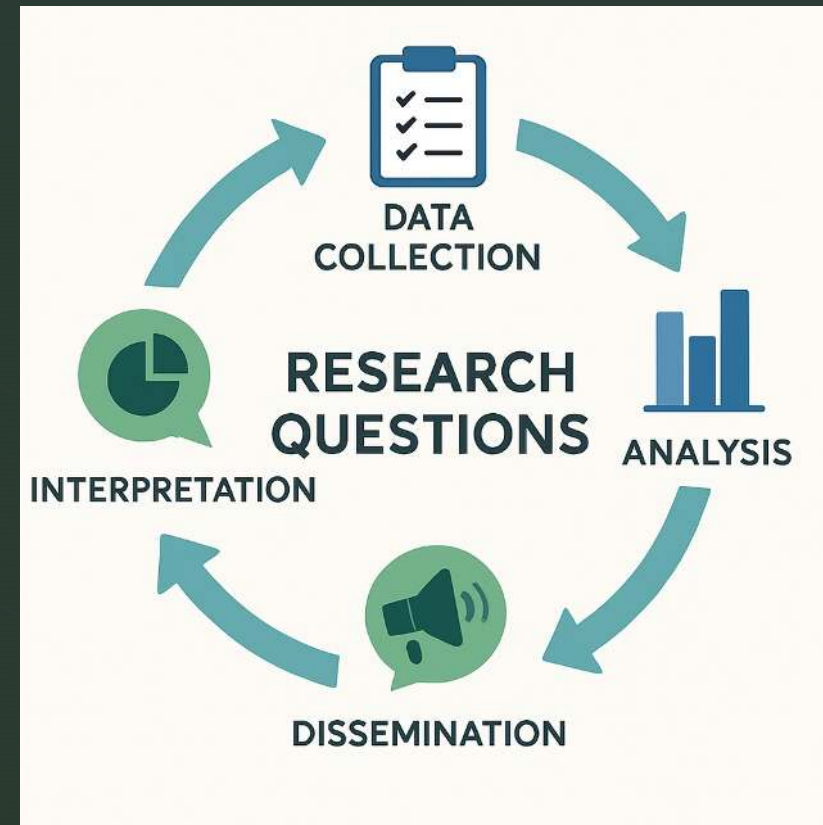


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R - Reusable

Maximizing the potential for future research and innovation:

- Clear licensing for data, code, and content
- Detailed provenance: know the origin and history of the data
- Enables replication and new discoveries

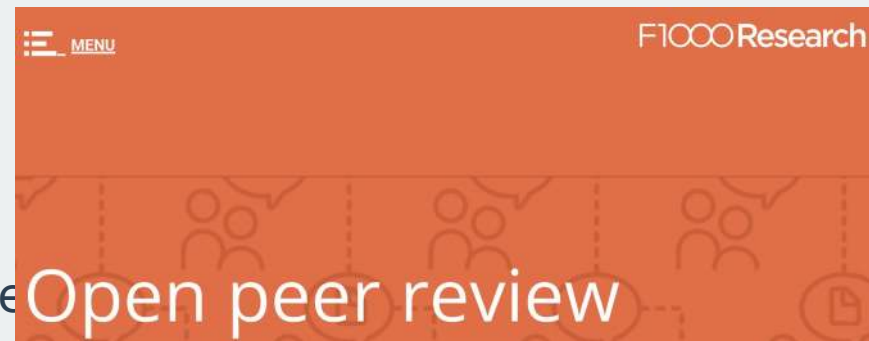


Applying the FAIR principles to peer review

Open peer review

-  **Findable:**
DOI for reviewers and reviews
create a discoverable record of the scientific discourse.
-  **Accessible:**
Publication of peer reviews alongside the article
Disclose the review model (e.g., single-blind, double-blind, open).
-  **Interoperable:**
Standardized formats for peer review across journals
Enable computational analysis of review feedback.
Facilitates meta-research.
-  **Reusable:**
Licenses that allow reutilization of comments
Detailed provenance: know the origin and history of the data.
-  **Benefits:**

Improved transparency, responsibility and recognition for reviewers



Applying the FAIR principles to Scientific Publication

FAIR publications:

Findable:

DOIs for articles and associated datasets

Accessible:

Open access or clear pathways to it

Interoperable:

Structured metadata for standardized formats

Reusable:

Explicit licenses (e.g.: CC-BY) and easy citation

Role of repositories:

Zenodo, Dryad, DataONE etc in ecology



^zBeyond FAIR Data: Towards FAIR & CARE

We must extend beyond FAIR to embrace the CARE Principles for ethical and equitable research, especially when it involves Indigenous knowledge and lands

CARE Principles for Indigenous Data Governance

Complementing the FAIR principles:

-  **C**ollective Benefit: Data that benefit indigenous communities
-  **A**uthority to Control: Recognition of governance rights upon the data
-  **R**esponsibility: Ethical and responsible use of indigenous data
-  **E**thics: Consideration of indigenous rights and wellbeing in all lifecycle of the data

Relevance to ecology:

Many field sites coincide with indigenous territories or use indigenous knowledge



Integrating FAIR and CARE in Peer Review and Publication



Application in Ecology:

 Inclusion of indigenous reviewers in ecological Research done in indigenous territories

 Acknowledgement of TEK as a contribution to scientific validation

 Protection of sensitive data about sacred places or (ethno)medicinal resources

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Reviewer Guidelines for CARE

Journals should provide guidelines for reviewers to assess:

- Evidence of FPIC from relevant Indigenous communities
- Plans for equitable benefit-sharing
- Respect for Indigenous data sovereignty and control
- Ethical considerations throughout the research

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Author Guidelines for CARE

Authors must clearly articulate how CARE principles have been addressed in their research:

- Include statements on consultation, consent, and benefit-sharing
- Acknowledge Indigenous knowledge holders and their contributions appropriately



Perspectives in ecology and conservation

Supported by Instituto Tecnológico Vale and CEPAN

www.perspectecolconserv.com



Policy Forums

Beyond data labor: sowing synthesis science in the Global South

André L. Luza^{a,*}, André L. Giles^b, Pedro J. Bergamo^c, Grasiela Casas^c, Alice R. de Moraes^c, Marina M. Monteiro^d, Herval V. Pinto-Junior^d, Kátia F. Rito^c, Ana C. Rorato^e, Lis F. Stegmann^f, Gisele R. Winck^g, Marisa Mamede^h



Discover Conservation

Perspective

Beyond more inclusive and equitable research in ecology and conservation: on the practical implications of expanding authorship to local data collectors

Ana Filipa Palmeirim¹

Benefits and Impacts

Benefits of the joint application FAIR + CARE:

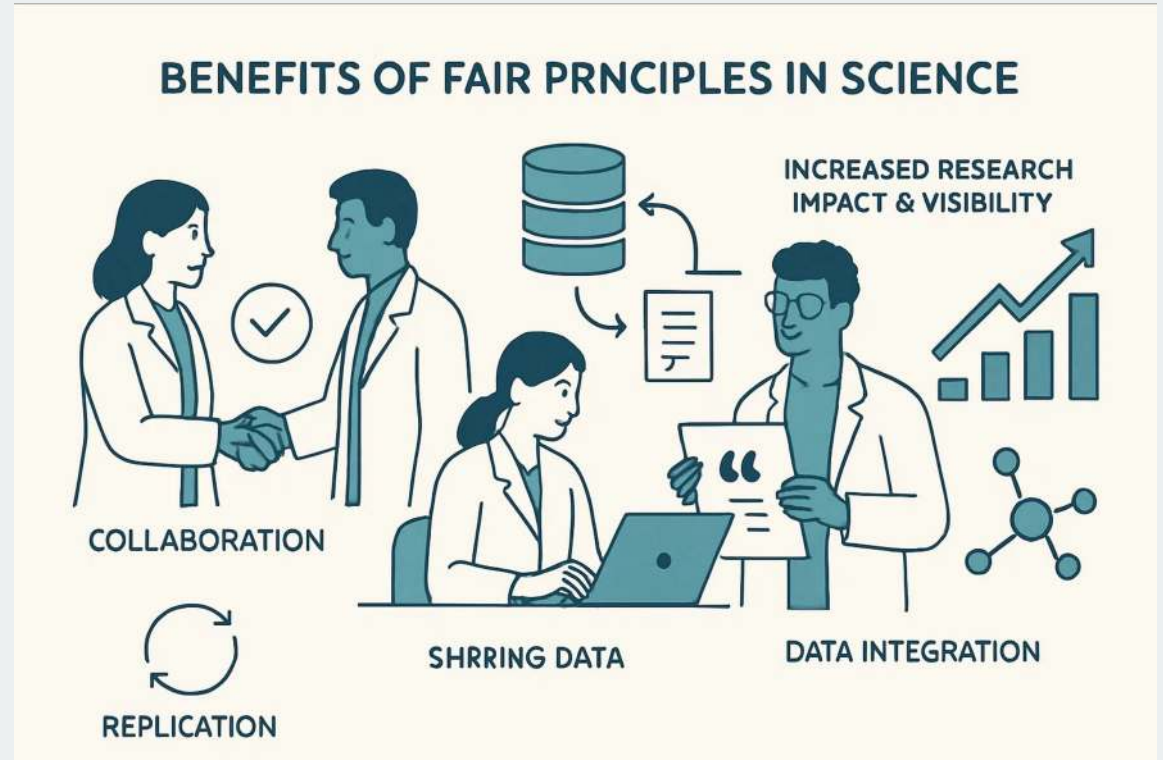
 **Enhanced Reproducibility**
with respect to traditional knowledge

 **More inclusive Collaboration**
Between scientists and indigenous communities

 **Higher impact and visibility**
For ecological research

 **Data Integration**
respecting indigenous rights and interests

 **More effective conservation**
Through the combination of western science
and traditional ecological knowledge



Challenges

Barriers

Cultural: Lack of awareness of CARE principles, navigating diverse Indigenous protocols, ensuring meaningful engagement of traditional peer review systems

Technical: Lack of infrastructure for complete implementation



Next Steps

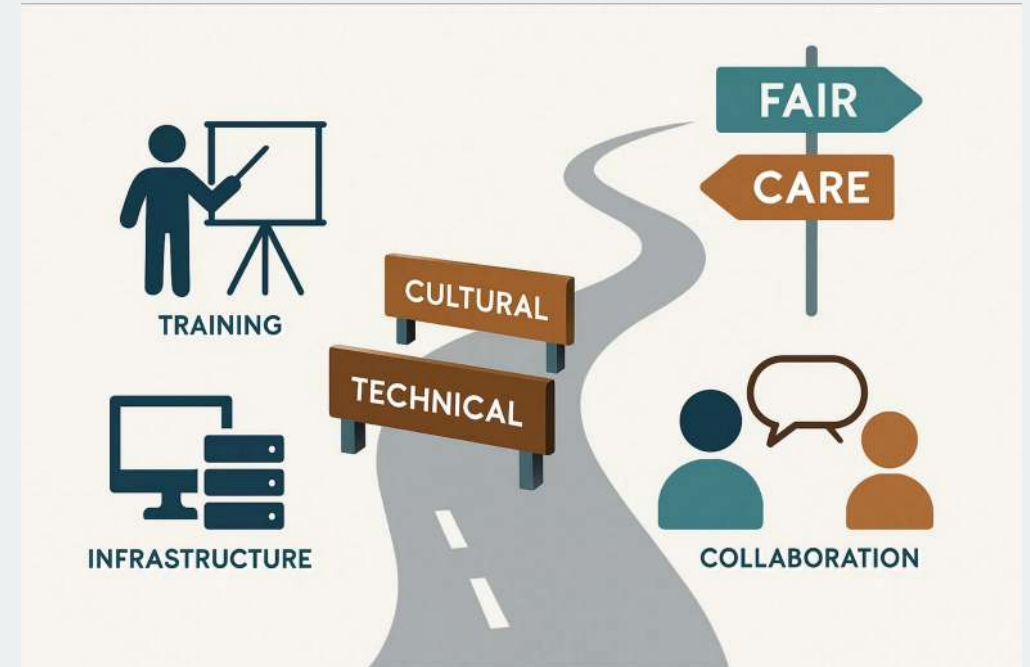
Next steps

Training

for editors, reviewers and researchers

Development of adequate **Infrastructure**

Collaboration between scientists and indigenous communities



Conclusion



✓ The **FAIR** and **CARE** principles complement each other and are essential for a more inclusive and efficient ecological science

✓ Ecology is an ideal field for the principles to be implemented due to its interdisciplinary nature and connection with indigenous territories/knowledge

✓ The integration of these principles into peer review and publication promotes transparency, reproducibility, and respect

✓ The future of ecology depends on an approach that is both technically robust and ethically responsible

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**Let's work together
to build a future for
ecological science
that is both FAIR and
For our publications,
our data, and our
review process.**

Thank you!

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